Delete, Retrieve, Generate: A Simple Approach to Sentiment and Style Transfer
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The study successfully illustrates how the confluence of novel, robust concepts from neural networks and language modeling helps implement style transfer – a mundane task in the light of the general nonalignment prevalent among sample sentences. Supplanting the deficiencies of the extant adversarial techniques with an elegant simplistic model, the results discussed in this work convince the reader that reduced complexity works in favor of enhanced comprehensibility. Yet simplification isn't always without demerits, and the model introduced in this report isn't an exception either. The task of identifying discriminative attribute markers was perhaps most challenging, considering the myriad of ways positive and negative sentiments are generally expressed. The definition of salience of n-grams introduced is crucial in this context and seems to be a perfect resolution at first sight; however, it isn't immune to logical infractions. Indeed, it isn't uncommon to find minimalistic sentences with a particular sentiment, driven by opposite or neutral polarity words. Sarcastic remarks or even idioms are perfect examples that fit this description. It could then be disputed how, despite such sentences' general negative/positive sentiment, n-grams of opposite polarity are captured as the attributes to be replaced. Indeed, substituting them with yet another opposite attribute, despite generating a new sentence, need not necessarily reverse the sentiment of the original one. If style transfer was the sole objective of this study, there needn't be much concern; however, with an additional sentiment reversal in mind, the model may have failed thus far. A similar contention could be found in cases where an otherwise positive or negative sentiment attribute markers play the role of content words. Sentences like "The good would have a different opinion" or "it is but not for the poor" having adjectival nouns are examples illustrating this development. Considering how 'good' or 'poor' are commonly used positive and negative sentiment attributes respectively, relying on a relative frequency-based salience definition could confuse the underlying neural network architecture in selecting the wrong attribute-marker from source sentences. In the absence of sufficient training data, reworking on the cited formulaic rules or revising a new strategy towards that end could quite possibly help control the occurrences of such discrepancies

Considering how grammatical sentence reconstruction is the predominant machinery driving style transfer, readers may be prompted to revisit the denoising autoencoder paradigms introduced in this discussion. Though the denoising methodology delivers promising results in this study, it may be too early to acknowledge its replicability across corpora. Sentence structures are admittedly varied. Thus, dovetailing random word attribute chunks with source sentences solely based on smoothing, helps choose other highly probable attribute words. Yet, these needn't necessarily yield error-free sentences since it may be the case that none of the other attributes selected after smoothing may still offer a satisfactory level of alignment or grammatical coherence with the original sentence. Thus, unless the target sentence is generic enough for a wide range of corpora, it goes without saying that the results may be unconventionally contingent on the training samples used. This subjectivity needn't always be a concern but nevertheless implies a logical flaw in some of the methodologies adopted in this research.